

The Program in Action

When you run *milliondicestats.py*, the output will look like this:

```
Million Dice Roll Statistics Simulator
By Al Sweigart al@inventwithpython.com

Enter how many six-sided dice you want to roll:
> 2
Simulating 1,000,000 rolls of 2 dice...
36.2% done...
73.4% done...
TOTAL - ROLLS - PERCENTAGE
 2 - 27590 rolls - 2.8%
 3 - 55730 rolls - 5.6%
 4 - 83517 rolls - 8.4%
 5 - 111526 rolls - 11.2%
 6 - 139015 rolls - 13.9%
 7 - 166327 rolls - 16.6%
 8 - 139477 rolls - 13.9%
 9 - 110268 rolls - 11.0%
10 - 83272 rolls - 8.3%
11 - 55255 rolls - 5.5%
12 - 28023 rolls - 2.8%
```

How It Works

We simulate the roll of a single six-sided die by calling `random.randint(1, 6)` on line 30. This returns a random number between 1 and 6, which gets added to the running total for however many dice are rolled together. The `random.randint()` function has a uniform distribution, meaning each number is just as likely as any other to be returned.

The program stores the results of this roll with the results dictionary. The keys to this dictionary are each possible dice roll total, and the values are how many times this total has been encountered. To get the frequency percentage, we divide the number of times a total has been encountered by 1,000,000 (the number of dice rolls in this simulation) and multiply it by 100 (to get a percentage between 0.0 and 100.0 instead of 0.0 and 1.0). By doing some algebra, we can figure out that this is the same as dividing the number of encounters by 10,000, which we do on line 37.

```
1. """Million Dice Roll Statistics Simulator
2. By Al Sweigart al@inventwithpython.com
3. A simulation of one million dice rolls.
4. View this code at https://nostarch.com/big-book-small-python-projects
5. Tags: tiny, beginner, math, simulation"""
6.
7. import random, time
8.
9. print('Million Dice Roll Statistics Simulator
10. By Al Sweigart al@inventwithpython.com
11.
```